Def. A Homotopy of Paths in X is a family

$$f_t : I \to X, \quad 0 \le t \le 1 \quad \text{such that}$$

1) For all $0 \le t \le 1$, $f_t(0) = x_0$ and $f_t(1) = x_1$.
2). $F : I \times I \to X ; (s,t) \mapsto f_t(s)$ is continuous.

Given two path p and q from $x_0 \in X$ to x_1, they are said to be homotopic if:

there exists a hopotopy of paths f_t satisfying $p = f_0$ and $q = f_1$.

Denoted as $p \simeq q$.

Example. Linear Homotopy

Given two path p and q in $\mathbb{R}^n$ having the same endpoint x_0 and x_1, define

$$f_t : I \to \mathbb{R}^n ; s \mapsto (1-t)p(s) + t\,q(s)$$

for each $0 \le t \le 1$. Then, given t,

$$f_t(0) = (1-t)p(0) + t\,q(0) = x_0 \quad \text{and} \quad f_t(1) = (1-t)p(1) + t\,q(1) = x_1$$

(In actually, for any $s \in [0,1]$, $p(s) = q(s) \Rightarrow f_t(s) = p(s)$)

And, to show $F : I \times I \to \mathbb{R}^n : (s,t) \mapsto f_t(s)$ is continuous, let $M > 0$ be a bound for p and q.

Let $(s_0, t_0) \in I \times I$ be given. Given $\varepsilon > 0$, there is $\delta > 0$ such that

$$\| s - s_0 \| < \delta \Rightarrow \| f_{t_0}(s) - f_{t_0}(s_0) \| < \varepsilon \text{ and}$$

$$\begin{aligned} \| t - t_0 \| < \delta \Rightarrow \| f_t(s) - f_{t_0}(s) \| &= \| (1-t)p(s) + t\,q(s) - [(1-t_0)p(s) + t_0\,q(s)] \| \\ &= \| (t_0 - t)p(s) + (t - t_0)q(s) \| \\ &\le |t_0 - t| \cdot (\| p(s) \| + \| q(s) \|) < \varepsilon. \end{aligned}$$

Now,

$$\| (s,t) - (s_0, t_0) \| < \delta \Rightarrow \| f_t(s) - f_{t_0}(s_0) \| \le \| f_t(s) - f_{t_0}(s) \| + \| f_{t_0}(s) - f_{t_0}(s_0) \| < 2\varepsilon.$$

So $p \simeq q$ ▫

Prop. Let X be a Topological Space. Given two Points x_0 and x_1 in X, define an relation on the Set of Paths from x_0 to x_1 as: $p \sim q \iff p \simeq q$.
Then, this relation is equivalence relation.
Proof. Reflexivity is evident, by $f_t = f$, the constant homotopy.
Symmetry is given by the inverse homotopy: if $f_0 \simeq f_1$ via f_t, then $f_1 \simeq f_0$ via f_{1-t}.
To show the Transivity, Suppose that $f_0 \simeq f_1$ via f_t and $g_0 \simeq g_1$ via g_t, with $f_1 = g_0$.
Define $H : I \times I \to X : (s,t) \mapsto \begin{cases} f_{2t}(s) & \text{if } 0 \le t \le \frac{1}{2} \\ g_{2t-1}(s) & \text{if } \frac{1}{2} \le t \le 1. \end{cases}$
It is well-defined since $f_1 = g_0$. And, Continuity is given by the Pasting Lemma.

Def. A path $p : I \to X$ is called a loop if $p(0) = p(1) = x_0 \in X$,
and the point x_0 is called the basepoint.
The Collection of all homotopy classes $[f]$ of loops $f : I \to X$ at the basepoint x_0,

$$\pi_1(X, x_0) \overset{\text{def}}{=} \{ [f] \mid f \text{ is a loop at } x_0 \}$$

is called the fundamental group under the operation

$$[f][g] \overset{\text{def}}{=} [f \cdot g] \text{ where } f \cdot g : I \to X : s \mapsto \begin{cases} f(2s) & \text{if } 0 \le s \le \frac{1}{2} \\ g(2s-1) & \text{if } \frac{1}{2} \le s \le 1 \end{cases}$$

This $\pi_1(X, x_0)$ is well-defined as a set clearly, and the operation is also well defined by:
Suppose that $[f_1] = [f_2]$ and $[g_1] = [g_2]$. Then $f_1 \simeq f_2$ and $g_1 \simeq g_2$ via $\hat{f}_t$ and $\hat{g}_t$.

$$[f_1][g_1] = [f_1 \cdot g_1] = [f_2 \cdot g_2] \text{ because } f_1 \cdot g_1 \simeq f_2 g_2 \text{ via } \hat{f}_t \cdot \hat{g}_t$$
$$= [f_2][g_2]$$

Moreover, the operation satisfies the Group Axiom, the Proof lies in the next Page.

proof of the fundamental group.

First, given a path $f: I \to X$, define a reparametrization of a path f as a composition

$$f \circ \varphi \quad \text{where } \varphi : I \to I \text{ is any continuous map s.t. } \varphi(0)=0 \text{ and } \varphi(1)=1.$$

The reparametrization preserves homotopy, that is,

$$f\circ\varphi \simeq f \text{ via } f\circ\varphi_t \text{ where } \varphi_t(s) = (1-t)\varphi(s) + ts, \text{ so that } \varphi_0 = \varphi \text{ and } \varphi_1 = s.$$

Associativity. Let f, g, h be paths with $f(1) = g(0)$ and $g(1) = h(0)$.

Then, the composition $(f\cdot g)\cdot h = f\cdot(g\cdot h)$ are defined. Moreover,

$f\cdot(g\cdot h)$ is a reparametrization of $(f\cdot g)\cdot h$, because

$$f\cdot(g\cdot h)(s) = \begin{cases} f(2s) & \text{if } 0 \le s \le \frac{1}{2} \\ g(4s-2) & \text{if } \frac{1}{2} \le s \le \frac{3}{4} \\ h(4s-3) & \text{if } \frac{3}{4} \le s \le 1 \end{cases} \quad \text{and}$$

$$(f\cdot g)\cdot h(s) = \begin{cases} f(4s) & \text{if } 0 \le s \le \frac{1}{4} \\ g(4s-1) & \text{if } \frac{1}{4} \le s \le \frac{1}{2} \\ h(2s-1) & \text{if } \frac{1}{2} \le s \le 1 \end{cases} \quad \text{therefore}$$

$$f\cdot(g\cdot h) = (f\cdot g)\cdot h \circ \varphi \quad \text{where} \quad \varphi(s) = \begin{cases} \frac{1}{2}s & \text{if } 0 \le s \le \frac{1}{2} \\ s - \frac{1}{4} & \text{if } \frac{1}{2} \le s \le \frac{3}{4} \\ 2s-1 & \text{if } \frac{3}{4} \le s \le 1 \end{cases}$$

So $f\cdot(g\cdot h) \simeq (f\cdot g)\cdot h$, the associative is proven

Identity. Given a path $f: I \to X$, let C be the constant path, as

$$C_i : I \to X ; s \mapsto f(i) \quad (i = 0, 1)$$

Then $f\cdot C_1$ is a reparametrization of f via

$$\varphi : I \to I ; s \mapsto \begin{cases} 2s & \text{if } 0 \le s \le \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \le s \le 1. \end{cases}$$

and similarly $C_0\cdot f$ is also a reparametrization, so, $C_0 f \simeq f\cdot C_1 \simeq f$.

For a loop, the endpoints are fixed, so $C(s) = x_0$ is two-sided identity in $\pi_1(X, x_0)$.

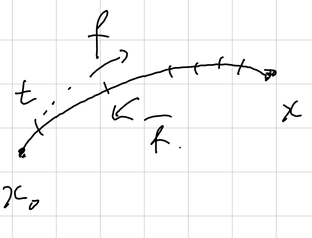
Inverse. Given a path f from x_0 to x_1, define the inverse path

$$\bar{f}(s) = f(1-s) \quad \text{so that } \bar{f}(0) = f(1),\ \bar{f}(1) = f(0)$$

Then $f\cdot\bar{f}$ is homotopic to a constant path $C(s) = f(0)$, via $h_t = f_t \cdot \bar{f}_t$ where

$$f_t : I \to X ; s \mapsto \begin{cases} f(s) & \text{if } 0 \le s \le 1-t \\ f(1-t) & \text{if } 1-t \le s \le 1 \end{cases}$$

similarly, for a loop, $f\cdot\bar{f} \simeq \bar{f}\cdot f \simeq C (\equiv x_0)$.



Change of basepoint.

Def. Let X be a space, and $x_0, x_1 \in X$ be points contained a same path-component.
Let $h: I \to X$ be a path from x_0 to x_1, with the inverse curve $\overline{h}(s) = h(1-s)$.
Then, for any loop at x_1, the composition $h \cdot f \cdot \overline{h}$ is a loop at x_0.
Rigorously, $(h \cdot f) \cdot \overline{h} \neq h \cdot (f \cdot \overline{h})$ but $(h \cdot f) \cdot \overline{h} \simeq h \cdot (f \cdot \overline{h})$. Therefore, the map

$$\beta_h : \pi_1(X, x_1) \to \pi_1(X, x_0) : [f] \mapsto [h \cdot f \cdot \overline{h}]$$

is well-defined, and is called a change of basepoint.
Moreover, for path-connected x_0 and x_1, the map β_h is an isomorphism.
Homomorphism: $\beta_h[f \cdot g] = [h \cdot f \cdot g \cdot \overline{h}] = [h \cdot f \cdot \overline{h} \cdot h \cdot g \cdot h] = \beta_h[f]\beta_h[g]$.

Invertible: $\beta_{\overline{h}}$ satisfies

$$\beta_h \beta_{\overline{h}}[f] = \beta_h[\overline{h} f h] = [h \overline{h} \cdot f \cdot h \cdot \overline{h}] = [f].$$

and similarly $\beta_{\overline{h}} \beta_h [f] = [f]$.

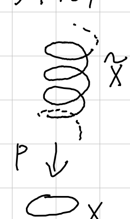
Def. A space X is called if it is path-connected and has trivial fundamental group.

Def. Given a space X, a covering space of X consists of a space $\tilde{X}$ and a map $p: \tilde{X} \to X$ s.t.
For each $x \in X$, there is an open neighborhood U of x in X satisfying

$$p^{-1}[U] = \bigsqcup_{i \in \Lambda} \mathcal{O}_i, \quad \text{each } \mathcal{O}_i \text{ s.t. } \mathcal{O}_i \overset{\text{Homeo}}{\cong} U \text{ via } p.$$

Example. Given a space S^1, let $p: \mathbb{R} \to S^1 : s \mapsto (\cos 2\pi s, \sin 2\pi s)$.
Let $(1,0) \in S^1$ be given. Then, for a neiborhood $B_1(1,0) \cap S_1$

$$p^{-1}[B_1(1,0) \cap S^1] = \bigsqcup_{n \in \mathbb{Z}} \left(n + \left(-\tfrac{1}{8}, \tfrac{1}{8}\right)\right)$$



and for each $n \in \mathbb{Z}$,

$$\left(n + \left(-\tfrac{1}{8}, \tfrac{1}{8}\right)\right) \cong B_1(1,0) \cap S^1 \text{ via } p.$$

┌ The Fundamental Group of the Circle.

┌ Thm. $\pi_1(S^1)$ is an infinite cyclic group generated by $[w]$, where

$$w(s) = (\cos 2\pi s, \sin 2\pi s) \text{ based at } (1,0).$$

Note that $[w]^n = [w_n]$ where $w_n(s) = (\cos 2\pi ns, \sin 2\pi ns)$ for each $n \in \mathbb{N}$.
Consider $p : \mathbb{R} \to S^1 ; s \mapsto (\cos 2\pi s, \sin 2\pi s)$ and $\tilde{w}_n : I \to \mathbb{R} ; s \mapsto ns$.
Then p is a covering map, and $\tilde{w}_n$ is called the lift of w_n, satisfying

$$w_n = p \circ \tilde{w}_n.$$

Proof. Let $f : I \to S^1$ be a loop at the basepoint $x_0 = (1,0)$.
Since S^1 is path-connected, $[f] \in \pi_1(S^1)$.